

HVAC Motor Anomaly Detection via Reconstruction-Based Acoustic Modeling and Live Microphone Inference

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1 Introduction

Traditional HVAC condition monitoring typically relies on vibration or temperature sensors, which require physical installation, increase deployment cost, and are difficult to scale in real-world building environments. In this project, we explore the use of acoustic sensing as a non-invasive and low-cost alternative. However, existing acoustic diagnostic methods often rely on computationally heavy offline analysis and high-fidelity equipment, leaving a gap between laboratory analysis and practical real-time deployment. To address this gap, the core innovation of our project is an edge-deployable, end-to-end acoustic anomaly detection workflow. This workflow combines reconstruction-based modeling, low-cost ESP32/I2S microphone hardware, and live microphone inference. Instead of merely conducting offline evaluations, we aim to demonstrate the feasibility of performing live inference directly at the edge, evaluating its advantages and limitations in practical applications.

2 Project Plan

Weeks 1–2 (03/09–03/15): Hardware Setup and Signal Stabilization

In the initial phase, we focused on building an acoustic sensing system using an ESP32 and an I2S microphone. We spent a significant amount of time resolving issues related to unstable signal output, including data gaps and invalid readings. By the end of this phase, we had successfully established a stable workflow for real-time audio acquisition, as well as an alternative approach using the microphone.

Week 3 (03/16–03/22): Preliminary Data Collection and Signal Exploration

We conducted preliminary data collection under various fan operating conditions to verify whether distinguishable patterns existed in the acoustic signals. Although the experimental setup was not yet fully controlled, these experiments confirmed that the collected data was meaningful.

Week 4 (03/23–03/30): Baseline Model and Demo System Development

We implemented a reconstruction-based anomaly detection model (an autoencoder using log-mel features) and built a simple demo system supporting uploaded audio and real-time microphone inference. The focus of this phase was to integrate data, models, and user interaction into a functional pipeline.

Current Phase (Week 5) (04/01–04/07): Problem Refinement and Structured Experimental Design

In this phase, our focus shifts from building the pipeline to conducting more structured experiments and clarifying the project’s evaluation strategy. Based on our current experimental plan, we begin

systematic data collection under a controlled 3×3 condition matrix: three operating conditions (normal, blocked airflow, and blade imbalance) under three voltage settings (4 V, 8 V, and 12 V). To reduce unnecessary variation, all recordings will be collected under a fixed setup, including a microphone distance of 15 cm and an angle of 45 degrees relative to the fan outlet.

For each condition, we plan to collect at least five recordings, resulting in a minimum of 45 recordings in total. These data will be used to support several comparisons: normal versus abnormal conditions, blocked airflow versus blade imbalance, the original baseline model versus a fine-tuned target-domain model, and offline uploaded-audio inference versus live microphone inference.

Our main evaluation metrics will include reconstruction MSE, threshold-based classification results, and score separation across conditions. In addition, we will examine qualitative consistency, such as whether similar physical conditions produce similar score patterns and whether live demo predictions broadly agree with offline analysis. The goal of this stage is not yet to claim final success, but to determine whether the current acoustic pipeline can produce meaningful and interpretable differences across controlled fan conditions.

Remaining Time: Evaluation and Optimization

Over the remaining 17 days, we plan to:

- Complete structured experiments and collect a consistent dataset
- Evaluate the model’s performance under different conditions
- Analyze system limitations and optimize system workflows
- Prepare final presentation and report materials

3 Challenges and Requested Assistance

Despite the progress made to date, we currently face several challenges that require further clarification and guidance, particularly as we recognize a lack of sufficient understanding regarding the project’s logic and scope. First, we are still gaining a deeper understanding of the latest advancements in the field of acoustic anomaly detection. In particular, we hope to gain a clearer understanding of how current methods differ from existing acoustic methods, as well as which performance metrics or evaluation criteria are considered meaningful in this field. This requires us to read more literature and engage in thorough discussions with our professor.

Second, although we have proposed a structured experimental plan, we believe there is room for further optimization. We are in the process of transitioning from exploratory experiments to more systematic testing and would appreciate feedback to confirm whether our current design—such as the condition matrix and recording strategy—is sufficiently rigorous.

Third, our evaluation framework has not yet been fully finalized. While we plan to use reconstruction error, threshold-based decision-making, and qualitative comparisons, guidance on how to design a clearer and more persuasive evaluation strategy would be highly beneficial. We plan to begin systematic testing in Week 5 and then present a preliminary evaluation proposal to the professor for discussion.

Additionally, we would like to receive feedback on the following aspects:

- Whether the current definition of “anomalies” is appropriate for the context of this project
- If the final deliverables should focus on experimental analysis, how the system development component should be revised and retained

Given the limited time remaining (approximately 17 days), we hope these discussions will help ensure the project remains focused and ultimately leads to clear and meaningful outcomes.